

5

Pressure and Horsepower Required

In previous chapters we discussed how to calculate friction factors and pressure loss due to friction in a pipeline using various equations such as Colebrook-White, Hazen-Williams, etc. We also analyzed the internal pressure allowable in a pipe and how to determine pipe wall thickness required for a specific internal pressure and pipe material, according to design code.

In this chapter we analyze how much pressure will be required at the beginning of a pipeline to safely transport a given throughput to the pipeline terminus, taking into account the pipeline elevation profile and the pipeline terminus pressure required, in addition to friction losses. We will also calculate the pumping horsepower required and in many cases also determine how many pump stations are needed to transport the specified volume of liquid.

We also examine pipes in series and parallel and how to calculate equivalent length of pipes in series and the equivalent diameter of parallel pipes. System head curves will be introduced along with flow injection and delivery along the pipeline. Incoming and outgoing branch connections will be studied as well as pipe loops.

5.1 Total Pressure Required

The total pressure P_T required at the beginning of a pipeline to transport a given flow rate from point A to point B will depend on

- Pipe diameter, wall thickness, and roughness
- Pipe length
- Pipeline elevation changes from A to B
- Liquid specific gravity and viscosity
- Flow rate

If we increase the pipe diameter, keeping all other items above constant, we know that the frictional pressure drop will decrease and hence the total pressure P_T will also decrease. Increasing pipe wall thickness or pipe roughness will cause increased frictional pressure drop and thus increase the value of P_T . On the other hand, if only the pipe length is increased, the pressure drop for the entire length of the pipeline will increase and so will the total pressure P_T .

How does the pipeline elevation profile affect P_T ? If the pipeline were laid in a flat terrain, with no appreciable elevation difference between the beginning of the pipeline A and the terminus B, the total pressure P_T will not be affected. But if the elevation difference between A and B were substantial, and B was at a higher elevation than A, P_T will be higher than that for the pipeline in flat terrain.

The higher the liquid specific gravity and viscosity, the higher will be the pressure drop due to friction and hence the larger the value of P_T . Finally, increasing the flow rate will result in a higher frictional pressure drop and therefore a higher value for P_T .

In general, the total pressure required can be divided into three main components as follows:

- Friction head
- Elevation head
- Delivery pressure at terminus

As an example, consider a pipeline from point A to point B operating at 4000 bbl/hr flow rate. If the total pressure drop due to friction in the pipeline is 800 psi, the elevation difference from point A to point B is 500 ft (uphill flow), and the minimum delivery pressure required at the terminus B is 50 psi, we can state that the pressure required at A is the sum of the three components as follows:

Total pressure at A = 800 psi + 500 ft + 50 psi

If the liquid specific gravity is 0.85, then using consistent units of psi the above equation reduces to

$$\text{Total pressure} = 800 + (500)(0.85/2.31) + 50 = 1033.98 \text{ psi}$$

Of course, this assumes that there are no controlling peaks or high elevation points between point A and point B. If an intermediate point C located halfway between A and B had an elevation of 1500 ft, compared with an elevation of 100 ft at point A and 600 ft at point B, then the elevation of point C becomes a controlling factor. In this case, the calculation of total pressure required at A is a bit more complicated.

Assume that the example pipeline is 50 miles long, of uniform diameter and thickness throughout its entire length, and the flow rate is 4000 bbl/hr. Therefore, the pressure drop per mile will be calculated as a constant value for the entire pipeline from the given values as

$$800/50 = 16 \text{ psi/mile}$$

The total frictional pressure drop between point A and the peak C located 25 miles away is

$$\text{Pressure drop from A to C} = 16 \times 25 = 400 \text{ psi}$$

Since C is the midpoint of the pipeline, an identical frictional pressure drop exists between C and B as follows:

$$\text{Pressure drop from C to B} = 16 \times 25 = 400 \text{ psi}$$

Now consider the portion of the pipeline from A to C, with a frictional pressure drop of 400 psi calculated above and an elevation difference between A and C of $1500 \text{ ft} - 100 \text{ ft} = 1400 \text{ ft}$. The total pressure required at A to get over the peak at C is the sum of the friction and elevation components as follows:

$$\text{Total pressure} = 400 + (1400)(0.85/2.31) = 915.15 \text{ psi}$$

It must be noted that this pressure of 915.15 psi at A will just about get the liquid over the peak at point C with zero gauge pressure. Sometimes it is desired that the liquid at the top of the hill be at some minimum pressure above the liquid vapor pressure at the flowing temperature. If the transported liquid were liquefied petroleum gas (LPG), we would require a minimum pressure in the pipeline of 250 to 300 psi. On the other hand with crude oils and refined products with low vapor pressure, the minimum pressure required may be only 10 to 20 psi. In this example, we assume that we are dealing with low vapor pressure liquids and that

therefore a minimum pressure of 10 psi is adequate at the high points in a pipeline. Our revised total pressure at A will then be

$$\text{Total pressure} = 400 + (1400)(0.85/2.31) + 10 = 925.15 \text{ psi}$$

Therefore, starting with a pressure of 925.15 psi at A will result in a pressure of 10 psi at the highest point C after accounting for the frictional pressure drop and the elevation difference between A and the peak C.

Once the liquid reaches the high point C at 10 psi, it flows downhill from point C to the terminus B at the given flow rate, being assisted by gravity. Therefore, for the section of the pipeline from C to B the elevation difference helps the flow while the friction impedes the flow. The arrival pressure at the terminus B can be calculated by considering the elevation difference and frictional pressure drop between C and B as follows:

$$\begin{aligned} \text{Delivery pressure at B} &= 10 + (1500 - 600)(0.85/2.31) - 400 \\ &= -58.83 \text{ psi} \end{aligned}$$

Since the calculated pressure at B is negative, it is clear that the specified minimum delivery pressure of 50 psi at B cannot be achieved with a starting pressure of 925.15 psi at A. Therefore, to provide the required minimum delivery pressure at B, the starting pressure at A must be increased to

$$\text{Pressure at A} = 925.15 + 50 + 58.83 = 1033.98 \text{ psi}$$

The above value is incidentally the same pressure we calculated at the beginning of this section, without considering the point C. Hence the revised pressure at peak C = $10 + 50 + 58.83 = 118.83$ psi.

Thus a pressure of 1033.98 psi at the beginning of the pipeline will result in a delivery pressure of 50 psi at B and clear the peak at C with a pressure of 118.83 psi. This is depicted in [Figure 5.1](#) where

$$P_A = 1034 \text{ psi}$$

$$P_C = 119 \text{ psi}$$

and

$$P_B = 50 \text{ psi}$$

All pressures have been rounded off to the nearest whole number.

Although the higher elevation point at C appeared to be controlling, calculation showed that the pressure required at A depended more on the required delivery pressure at the terminus B. In many cases this may not be true. The pressure required will be dictated by the controlling peak and therefore the arrival pressure of the liquid at the pipeline terminus may be higher than the minimum required.

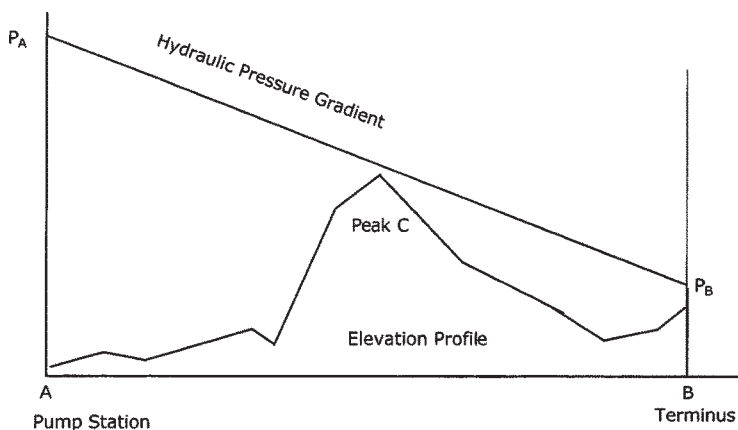


Figure 5.1 Hydraulic gradient.

Consider a second example in which the intermediate peak will be a controlling factor. Suppose the above pipeline now operates at 2200 bbl/hr and the pressure drop due to friction is calculated to be 5 psi/mile.

We will first calculate the pressure required at A, ignoring the peak elevation at C. The pressure required at A is

$$5 \times 50 + (600 - 100) \times 0.85 / 2.31 + 50 = 484 \text{ psi}$$

Based on this 484 psi pressure at A, the pressure at the highest point C will be calculated by deducting from 484 psi the pressure drop due to friction from A to C and adjusting for the elevation increase from A to C as follows:

$$P_C = 484 - (5 \times 25) - (1500 - 100) \times 0.85 / 2.31 = -156 \text{ psi}$$

This negative pressure is not acceptable, since we require a minimum positive pressure of 10 psi at the peak to prevent liquid vaporization. It is therefore clear that the pressure at A calculated above is inadequate. The controlling peak at C therefore dictates the pressure required at A. We will now calculate the revised pressure at A to maintain a positive pressure of 10 psi at the peak at C:

$$P_A = 484 + 156 + 10 = 650 \text{ psi}$$

Therefore, starting with a pressure of 650 psi at A provides the required minimum of 10 psi at the peak C. The delivery pressure at B can now be calculated as

$$P_B = 650 - (5 \times 50) - (600 - 100) \times 0.85 / 2.31 = 216 \text{ psi}$$

which is more than the required minimum terminus pressure of 50 psi.

This example illustrates the approach used in considering all critical elevation points along the pipeline to determine the pressure required to transport the liquid.

Next we repeat this analysis for a higher vapor pressure product.

If we were pumping a high vapor pressure liquid that requires a delivery pressure of 500 psi at the terminus, we would calculate the required pressure at A as follows:

$$\text{Pressure at A} = (6 \times 50) + \frac{(500 \times 0.65)}{2.31} + 500 = 940.69 \text{ psi}$$

where the high vapor pressure liquid ($S_g=0.65$) is assumed to produce a pressure drop of 6 psi/mile for the same flow rate. The pressure at the peak will be:

$$\text{Pressure at C} = 940.69 - (6 \times 25) - (1500 - 100) \times \frac{0.65}{2.31} = 396.75 \text{ psi}$$

If the higher vapor pressure product requires a minimum pressure of 400 psi, it can be seen from above that we do not have adequate pressure at the peak C. We therefore need to increase the starting pressure at A to

$$940.69 + (400 - 396.75) = 944 \text{ psi, rounded off}$$

With this change, the delivery pressure at the terminus B will then be

$$500 + (944 - 940.69) = 503 \text{ psi}$$

This is illustrated in [Figure 5.2](#).

5.2 Hydraulic Pressure Gradient

Generally, due to friction losses the liquid pressure in a pipeline decreases continuously from the pipe inlet to the pipe delivery terminus. If there is no elevation difference between the two ends of the pipeline and the pipe elevation profile is essentially flat, the inlet pressure at the beginning of the pipeline will decrease continuously by the friction loss at a particular flow rate. When there are elevation differences along the pipeline, the decrease in pipeline pressure along the pipeline will be due to the combined effect of pressure drop due to friction and the algebraic sum of pipeline elevations. Thus with a starting pressure of 1000 psi at the beginning of the pipeline, and assuming 15 psi/mile pressure drop due to friction in a flat

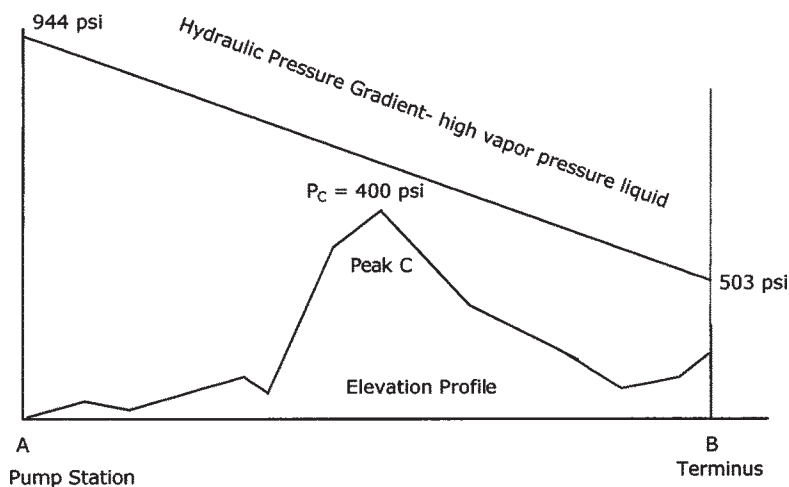


Figure 5.2 Hydraulic gradient: high vapor pressure liquid.

pipeline (no elevation difference) with constant diameter, the pressure at a distance of 20 miles from the beginning of the pipeline would drop to

$$1000 - 15 \times 20 = 700 \text{ psi}$$

If the pipeline is 60 miles long, the pressure drop due to friction in the entire line will be

$$15 \times 60 = 900 \text{ psi}$$

The pressure at the end of the pipeline will be

$$1000 - 900 = 100 \text{ psi}$$

Thus, the liquid pressure in the pipeline has uniformly dropped from 1000 psi at the beginning of the pipeline to 100 psi at the end of the 60 mile length. This pressure profile is referred to as the hydraulic pressure gradient in the pipeline. The hydraulic pressure gradient is a graphical representation of the variation in pressure along the pipeline. It is shown along with the pipeline elevation profile. Since elevation is plotted in feet, it is convenient to represent the pipeline pressures also in feet of liquid head. This is shown in [Figures 5.1–5.3](#).

In the example discussed in Section 5.1, we calculated the pressure required at the beginning of the pipeline to be 1034 psi for pumping crude oil at a flow rate of 4000 bbl/hr. This pressure requires one pump station at

the origin of the pipeline (point A). Assume now that the pipe length is 100 miles and the maximum allowable operating pressure (MAOP) is limited to 1200 psi. Suppose the total pressure required at A is calculated to be 1600 psi at a flow rate of 4000 bbl/hr. We would then require an intermediate pump station between A and B to limit the maximum pressure in the pipeline to 1200 psi. Due to the MAOP limit, the total pressure required at A will be provided in steps. The first pump station at A will provide approximately half the pressure while a second pump station located at some intermediate point provides the other half. This results in a sawtooth-like hydraulic gradient as shown in Figure 5.4. The actual discharge pressures at each pump station will be calculated considering pipeline elevations between A and B and the required minimum suction pressures at each of the two pump stations. An approximate calculation is described below, referring to Figure 5.4.

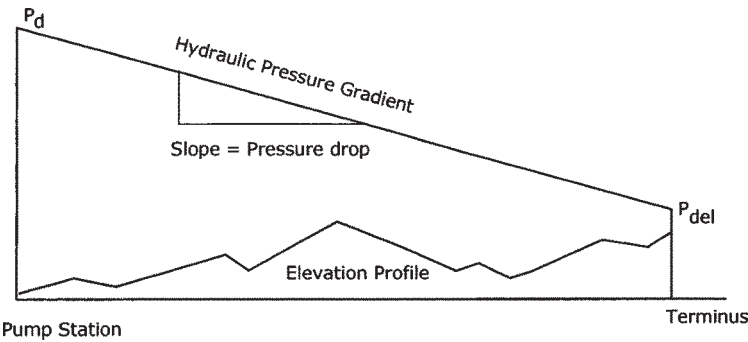


Figure 5.3 Hydraulic pressure gradient.

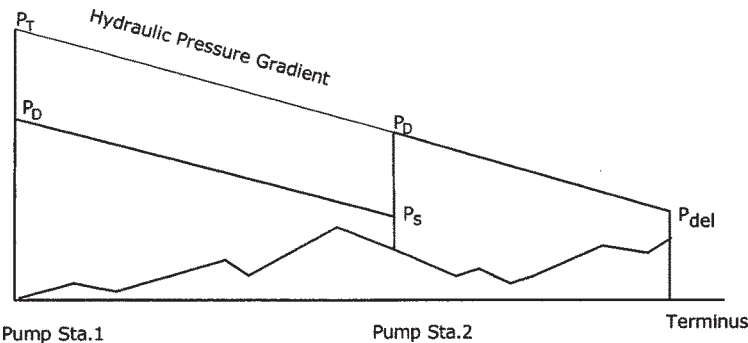


Figure 5.4 Hydraulic pressure gradient: two pump stations.

Let P_s and P_d represent the common suction and discharge pressure, respectively, for each pump station, while P_{del} is the required delivery pressure at the pipe terminus B. The total pressure P_t required at A can be written as follows.

$$P_t = P_{friction} + P_{elevation} + P_{del} \quad (5.1)$$

where

P_t = Total pressure required at A

$P_{friction}$ = Total frictional pressure drop between A and B

$P_{elevation}$ = Elevation head between A and B

P_{del} = Required delivery pressure at B

Also, from [Figure 5.4](#), based on geometry, we can state that

$$P_t = P_d + P_d - P_s \quad (5.2)$$

Solving for P_d we get

$$P_d = (P_t + P_s) / 2 \quad (5.3)$$

where

P_d = Pump station discharge pressure

P_s = Pump station suction pressure

For example, if the total pressure calculated is 1600 psi and the MAOP is 1200 psi we would need two pump stations. Considering a minimum suction pressure of 50 psi, each pump station would have a discharge pressure of

$$P_d = (1600 + 50) / 2 = 825 \text{ psi using Equation (5.3)}$$

Each pump station operates at 825 psi discharge pressure and the pipeline MAOP is 1200 psi. It is clear that based on pipeline pressures alone we have the capability of increasing pipeline throughput further to fully utilize the 1200 psi MAOP at each pump station. Of course, this would correspondingly require enhancing the pumping equipment at each pump station, since more horsepower (HP) will be required at the higher flow rate. We can now estimate the increased throughput possible if we were to operate the pipeline at the 1200 psi MAOP level at each pump station.

Let us assume that the pipeline elevation difference in this example contributes 300 psi to the total pressure required. This simply represents the station elevation head between A and B converted to psi. This component of the total pressure required ($P_t = 1600$ psi) depends only on the pipeline elevation and liquid specific gravity and therefore does not vary with flow rate. Similarly, the delivery pressure of 50 psi at B is also

independent of flow rate. We can then calculate the frictional component (which depends on flow rate) of the total pressure P_t using Equation 5.1 as follows:

$$\text{Frictional pressure drop} = 1600 - 300 - 50 = 1250 \text{ psi}$$

Assuming a pipeline length of 100 miles, the frictional pressure drop per mile of pipe is

$$P_m = 1250 / 100 = 12.5 \text{ psi/mile}$$

This pressure drop occurs at a flow rate of 4000 bbl/hr. From [Chapter 3](#) on pressure drop calculations, we know that the pressure drop per mile, P_m , varies as the square of the flow rate as long as the liquid properties and pipe size do not change. Using Equation (3.28) we can write

$$P_m = K(Q)^2 \quad (5.4)$$

where

P_m = Frictional pressure drop per mile of pipe

K = A constant for this pipeline that depends on liquid properties and pipe diameter

Q = Pipeline flow rate

Note that the K value above is not the same as the head loss coefficient discussed in Chapter 3. Strictly speaking, K also includes a transmission factor F (or a friction factor f) which varies with flow rate. However, for simplicity, we will assume that K is the constant that encompasses liquid specific gravity and pipe diameter. A more rigorous approach requires an additional parameter in Equation (5.4) that would include the transmission factor F , which in turn depends on Reynolds number, pipe roughness, etc.

Therefore, using Equation (5.4) we can write for the initial flow rate of 4000 bbl/hr

$$12.5 = K(4000)^2 \quad (5.5)$$

In a similar manner, we can estimate the frictional pressure drop per mile when flow rate is increased to some value Q to fully utilize the 1200 psi MAOP of the pipeline.

Using Equation (5.3), if we allow each pump station to operate at 1200 psi discharge pressure, we can write

$$1200 = (P_i + 50) / 2$$

or

$$P_i = 2400 - 50 = 2350 \text{ psi}$$

This total pressure will now consist of friction, elevation, and delivery pressure components at the higher flow rate Q . From Equation (5.1) we can write

$$2350 = P_{\text{friction}} + P_{\text{elevation}} + P_{\text{del}} \quad \text{at the higher flow rate } Q$$

or

$$2350 = P_{\text{friction}} + 300 + 50$$

Therefore,

$$P_{\text{friction}} = 2350 - 300 - 50 = 2000 \text{ psi at the higher flow rate } Q$$

Thus the pressure drop per mile at the higher flow rate Q is

$$P_m = 2000/100 = 20 \text{ psi/mile}$$

From Equation (5.4) we can write

$$20 = K(Q)^2 \tag{5.6}$$

where Q is the unknown higher flow rate in bbl/hr.

By dividing Equation (5.6) by Equation (5.5) we get the following:

$$20/12.5 = (Q/4000)^2$$

Solving for Q we get

$$Q = 4000(20/12.5)^{1/2} = 5059.64 \text{ bbl/hr}$$

Therefore, by fully utilizing the 1200 psi MAOP of the pipeline with the two pump stations, we are able to increase the flow rate to approximately 5060 bbl/hr. As previously mentioned, this will definitely require additional pumps at both pump stations to provide the higher discharge pressure. Pumps will be discussed in [Chapter 7](#).

In the preceding sections we have considered a pipeline of uniform diameter and wall thickness for its entire length. In reality, pipe diameter and wall thickness change, depending on the service requirement, design code, and local regulatory requirements. Pipe wall thickness may have to be increased due to differences in the specified minimum yield strength (SMYS) of pipe because a higher or lower grade of pipe was used at some locations. As mentioned previously, some cities or counties through which the pipeline traverses may require different design factors (0.66 instead of 0.72) to be used, thus necessitating a different wall thickness. If there are drastic elevation changes along the pipeline, the low elevation points may require higher wall thickness to withstand the higher pipe operating

pressures. If the pipeline has intermediate flow delivery or injections, the pipe diameter may be reduced or increased for certain portions to optimize pipe use. In all these cases, we can conclude that the pressure drop due to friction will not be uniform throughout the entire pipeline length. Injections and deliveries along the pipeline and their impact on pressure required are discussed later in this chapter.

When pipe diameter and wall thickness change along a pipeline, the slope of the hydraulic gradient, as shown in [Figure 5.3](#), will no longer be uniform. Due to varying frictional pressure drop (because of changes in pipe diameter and wall thickness), the slope of the hydraulic gradient will vary along the pipe length.

5.3 Series Piping

Pipes are said to be in series if different lengths of pipes are joined end to end with the entire flow passing through all pipes, without any branching.

Consider a pipeline consisting of two different lengths and pipe diameters joined together in series. A pipeline 1000 ft long and 16 in. in diameter connected in series with a pipeline 500 ft long and 14 in. in diameter would be an example of a series pipeline. At the connection point we will need to have a fitting, known as a reducer, that will join the 16 in. pipe with the smaller 14 in. pipe. This fitting will be a 16 in.×14 in. reducer. The reducer causes transition in the pipe diameter smoothly from 16 in. to 14 in. We can calculate the total pressure drop through this 16 in./14 in. pipeline system by adding the individual pressure drops in the 16 in. and the 14 in. pipe segments and accounting for the pressure loss in the 16 in.×14 in. reducer.

If two pipes of different diameters are connected together in series, we can also use the equivalent-length approach to calculate the pressure drop in the pipeline as discussed next.

A pipe is equivalent to another pipe or pipeline system when the same pressure loss due to friction occurs in the first pipe compared with that in the other pipe or pipeline system. Since the pressure drop can be caused by an infinite combination of pipe diameter and pipe length, we must specify a particular diameter to calculate the equivalent length.

Suppose a pipe A of length L_A and internal diameter D_A is connected in series with a pipe B of length L_B and internal diameter D_B . If we were to replace this two-pipe system with a single pipe of length L_E and diameter D_E , we have what is known as the equivalent length of pipe. This equivalent length of pipe may be based on one of the two diameters (D_A or D_B) or a totally different diameter D_E .

The equivalent length L_E in terms of pipe diameter D_E can be written as

$$L_E/(D_E)^5 = L_A/(D_A)^5 + L_B/(D_B)^5 \quad (5.7)$$

This formula for equivalent length is based on the premise that the total friction loss in the two-pipe system exactly equals that in the single equivalent pipe.

Equation (5.7) is based on Equation (3.28) for pressure drop, since pressure drop per unit length is inversely proportional to the fifth power of the diameter. If we refer to the diameter D_A as the basis, this equation becomes, after setting $D_E = D_A$,

$$L_E = L_A + L_B(D_A/D_B)^5 \quad (5.8)$$

Thus, we have an equivalent length L_E that will be based on diameter D_A . This length L_E of pipe diameter D_A will produce the same amount of frictional pressure drop as the two lengths L_A and L_B in series. We have thus simplified the problem by reducing it to one single pipe length of uniform diameter D_A .

It must be noted that the equivalent-length method discussed above is only approximate. Furthermore, if elevation changes are involved it becomes more complicated, unless there are no controlling elevations along the pipeline system.

An example will illustrate this concept of equivalent pipe length. Consider a 16 in. pipeline of 0.281 in. wall thickness and 20 miles long installed in series with a 14 in. pipeline of 0.250 in. wall thickness and 10 miles long. The equivalent length of this pipeline, using Equation (5.8), is

$$20 + 10 \times (16/14)^5 / (14/0.50)^5$$

$$= 39.56 \text{ miles of 16 in. pipe.}$$

The actual physical length of 30 miles of 16 in. and 14 in. pipes is replaced, for the purpose of pressure drop calculations with a single 16 in. pipe 39.56 miles long. Note that we have left out the pipe fitting that would connect the 16 in. pipe with the 14 in. pipe. This would be a 16 in. $\times$ 14 in. reducer which would have its own equivalent length. To be precise, we should determine the equivalent length of the reducer from [Table A.10](#) in Appendix A and add it to the above length to obtain the total equivalent length, including the fitting.

Once an equivalent pipe length has been determined, we can calculate the pressure drop based on this pipe size.

5.4 Parallel Piping

Pipes are said to be in parallel if they are connected in such a way that the liquid flow splits into two or more separate pipes and rejoins downstream into another pipe as illustrated in Figure 5.5. In Figure 5.5 liquid flows through pipe AB until at point B part of the flow branches off into pipe BCE, while the remainder flows through pipe BDE. At point E, the flows recombine to the original value and the liquid flows through the pipe EF. Note that we are assuming that all pipes in Figure 5.5 are shown in plan view with no elevation changes.

In order to calculate the pressures and flow rates in a parallel piping system such as the one depicted in Figure 5.5, we use the following two principles of pipes in parallel:

1. Conservation of total flow
2. Common pressure loss across each parallel pipe

According to principle 1, the total flow entering each junction of pipe must equal the total flow leaving the junction, or, simply,

$$\text{Total inflow} = \text{Total outflow}$$

Thus, in Figure 5.5, all flows entering and leaving the junction B must satisfy the above principle. If the flow into the junction B is Q and the flow in branch BCE is Q_{BC} and flow in the branch BDE is Q_{BD} , we have from the above conservation of total flow:

$$Q = Q_{BC} + Q_{BD} \quad (5.9)$$

assuming Q_{BC} and Q_{BD} represent the flow out of the junction B.

The second principle of parallel pipes, defined as principle 2 above, requires that the pressure drop across the branch BCE must equal the pressure drop across the branch BDE. This is simply due to the fact that point B represents the common upstream pressure for each of these

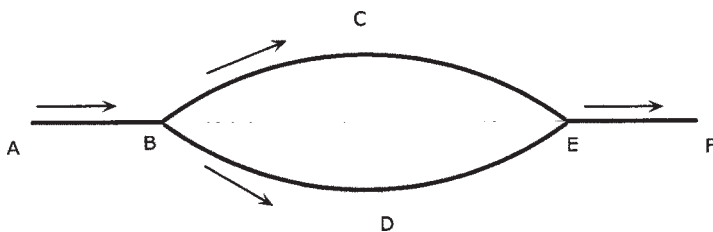


Figure 5.5 Parallel pipes.

branches, while the pressure at point E represents the common downstream pressure. Referring to these pressures as P_B and P_E , we can state

$$\text{Pressure drop in branch BCE} = P_B - P_E \quad (5.10)$$

$$\text{Pressure drop in branch BDE} = P_B - P_E \quad (5.11)$$

assuming that the flows Q_{BC} and Q_{BD} are in the direction of BCE and BDE respectively. If we had a third pipe branch between B and E, such as that shown by the dashed line BE in Figure 5.5, we can state that the common pressure drop $P_B - P_E$ would also be applicable to the third parallel pipe between B and E, as well.

We can rewrite the Equations (5.9) and (5.11) above as follows for the system with three parallel pipes:

$$Q = Q_{BC} + Q_{BD} + Q_{BE} \quad (5.12)$$

and

$$\Delta P_{BCE} = \Delta P_{BDE} = \Delta P_{BE} \quad (5.13)$$

where

ΔP = Pressure drop in respective parallel pipes.

Using the Equations (5.12) and (5.13) we can solve for flow rates and pressures in any parallel piping system. We will demonstrate the above using an example problem later in this chapter (Example Problem 5.1).

Similar to the equivalent-length concept in series piping, we can calculate an equivalent pipe diameter for pipes connected in parallel. Since each of the parallel pipes in Figure 5.5 has a common pressure drop indicated by Equation (5.13), we can replace all the parallel pipes between B and E with one single pipe of length L_E and diameter D_E such that the pressure drop through the single pipe at flow Q equals that of the individual pipes as follows:

Pressure drop in equivalent single pipe length L_E
and diameter D_E at flow rate $Q = \Delta P_{BCE}$

Assuming now that we have only the two parallel pipes BCE and BDE in Figure 5.5, ignoring the dashed line BE, we can state that

$$Q = Q_{BC} + Q_{BD}$$

and

$$\Delta P_{EQ} = \Delta P_{BCE} = \Delta P_{BDE}$$

Using Equation (3.28) the pressure ΔP_{EQ} for the equivalent pipe can be written as

$$\Delta P_{EQ} = K(L_E)(Q)^2/D_E^5$$

where K is a constant that depends on the liquid properties. The above two equations will then become

$$KL_E Q^2/D_E^5 = KL_{BC} Q_{BC}^2/D_{BC}^5 = KL_{BD} Q_{BD}^2/D_{BD}^5$$

Simplifying we get

$$L_E Q^2/D_E^5 = L_{BC} Q_{BC}^2/D_{BC}^5 = L_{BD} Q_{BD}^2/D_{BD}^5$$

Further simplifying the problem by setting

$$L_{BC}=L_{BD}=L_E$$

we get

$$Q^2/D_E^5 = Q_{BC}^2/D_{BC}^5 = Q_{BD}^2/D_{BD}^5$$

Substituting for Q_{BD} in terms for Q_{BC} from Equation (5.9) we get

$$Q^2/D_E^5 = Q_{BC}^2/D_{BC}^5 \quad (5.14)$$

and

$$Q_{BC}^2/D_{BC}^5 = (Q - Q_{BC})^2/D_{BD}^5 \quad (5.15)$$

From Equations (5.14) and (5.15) we can solve for the two flows Q_{BC} and Q_{BD} , and the equivalent diameter D_E , in terms of the known quantities Q , D_{BC} , and D_{BD} .

A numerical example will illustrate the above method.

Example Problem 5.1

A parallel pipe system, similar to the one shown in [Figure 5.5](#), is located in a horizontal plane with the following data:

Flow rate $Q=2000$ gal/min of water

Pipe branch BCE=12 in. diameter, 8000 ft

Pipe branch BDE=10 in. diameter, 6500 ft

Calculate the flow rate through each parallel pipe and the equivalent pipe diameter for a single pipe 5000 ft long between B and E to replace the two parallel pipes.

Solution

$$Q_1 + Q_2 = 2000$$

$$Q_1^2 L_1 / D_1^5 = Q_2^2 L_2 / D_2^5$$

where suffixes 1 and 2 refer to the two branches BCE and BDE respectively.

$$\begin{aligned} (Q_2/Q_1)^2 &= (D_2/D_1)^5 (L_1/L_2) \\ &= (10/12)^5 \times (8000/6500) \end{aligned}$$

$$Q_2/Q_1 = 0.7033$$

Solving we get

$$Q_1 = 1174 \text{ gal/min}$$

$$Q_2 = 826 \text{ gal/min}$$

The equivalent pipe diameter for a single pipe 5000 ft long is calculated as follows:

$$(2000)^2 (5000) / D_E^5 = (1174)^2 \times 8000 / (12)^5$$

or

$$D_E = 13.52 \text{ in.}$$

Therefore, a 13.52 in. diameter pipe, 5000 ft long, between B and E will replace the two parallel pipes.

5.5 Transporting High Vapor Pressure Liquids

As mentioned previously, transportation of high vapor pressure liquids such as liquefied petroleum gas (LPG) requires that a certain minimum pressure be maintained throughout the pipeline. This minimum pressure must be greater than the liquid vapor pressure at the flowing temperature, otherwise liquid may vaporize causing two-phase flow in the pipeline which the pumps cannot handle. If the vapor pressure of LPG at the flowing temperature is 250 psi, the minimum pressure anywhere in the pipeline must be greater than 250 psi. Conservatively, at high elevation points or peaks along the pipeline, we must insure that more than the minimum pressure is maintained. This is illustrated in [Figure 5.2](#). Additionally, the delivery pressure at the end of the pipeline must also satisfy the minimum pressure requirements. Thus, the delivery pressure at the pipeline terminus for LPG may be 300 psi or higher to account for any

meter station and manifold piping losses at the delivery point. Also, sometimes with high vapor pressure liquids the delivery point may be a pressure vessel or a pressurized sphere maintained at 500 to 600 psi and therefore may require even higher minimum pressures compared with the vapor pressure of the liquid. Hence, both the delivery pressure and the minimum pressure must be considered when analyzing pipelines transporting high vapor pressure liquids.

5.6 Horsepower Required

So far we have examined the pressure required to transport a given amount of liquid through a pipeline system. Depending on the flow rate and MAOP of the pipeline, we may need one or more pump stations to safely transport the specified throughput. The pressure required at each pump station will generally be provided by centrifugal or positive displacement pumps. Pump operation and performance will be discussed in [Chapter 7](#). In this section we calculate the horsepower required to pump a given volume of liquid through the pipeline regardless of the type of pumping equipment used.

5.6.1 Hydraulic Horsepower

Power required is defined as energy or work performed per unit time. In English units, energy is measured in foot pounds (ft·lb) and power is expressed in horsepower (HP). One HP is defined as 33,000 ft·lb/min or 550 ft·lb/s. In SI units, energy is measured in joules and power is measured in joules/second (watts). The larger unit kilowatt (kW) is more commonly used. One HP is equal to 0.746 kW.

To illustrate the concept of work, energy, and power required, imagine a situation that requires 150,000 gal of water to be raised 500 ft to supply the needs of a small community. If this requirement is on a 24 hr basis we can state that the work done in lifting 150,000 gal of water by 500 ft is

$$(150,000/7.48) \times 62.34 \times 500 = 625,066,845 \text{ ft·lb}$$

where the specific weight of water is assumed to be 62.34 lb/ft³ and 1 ft³=7.48 gal. Thus we need to expend 6.25×10^8 ft·lb of energy over a 24 hr period to accomplish this task. Since 1 HP equals 33,000 ft·lb/min, the power required in this case is

$$\text{HP} = \frac{6.25 \times 10^8}{24 \times 60 \times 33,000} = 13.2$$

This is also known as the hydraulic horsepower (HHP), since we have not considered pumping efficiency.

As a liquid flows through a pipeline, pressure loss occurs due to friction. The pressure needed at the beginning of the pipeline to account for friction and any elevation changes is then used to calculate the amount of energy required to transport the liquid. Factoring in the time element, we get the power required to transport the liquid.

Example Problem 5.2

Consider 4000 bbl/hr being transported through a pipeline with one pump station operating at 1000 psi discharge pressure. If the pump station suction pressure is 50 psi, the pump has to produce 1000-50 psi, or 950 psi differential pressure to pump 4000 bbl/hr of the liquid. If the liquid specific gravity is 0.85 at flowing temperature, calculate the HP required at this flow rate.

Solution

Flow rate of liquid in lb/min is calculated as follows:

$$M = 4000 \text{ bbl/hr} (5.6146 \text{ ft}^3/\text{bbl})(1 \text{ hr}/60 \text{ min})(0.85)(62.34 \text{ lb}/\text{ft}^3)$$

where 62.34 is the specific weight of water in lb/ft.³

or

$$M = 19,834.14 \text{ lb/min}$$

Therefore the HP required is

$$\text{HP} = \frac{(\text{lb/min})(\text{ft head})}{33,000}$$

or

$$\text{HP} = [(19,834.14)(950)(2.31)/0.85]/33,000 = 1552$$

It must be noted that in the above calculation no efficiency value has been considered. In other words, we have assumed 100% pumping efficiency. Therefore, the above HP calculated is referred to as hydraulic horsepower (HHP), based on 100% efficiency.

$$\text{HHP} = 1552$$

5.6.2 Brake Horsepower

The brake horsepower takes into account the pump efficiency. If a pump efficiency of 75% is used we can calculate the brake horsepower (BHP) in

the above example as follows:

Brake horsepower=Hydraulic horsepower/Pump efficiency

$$\text{BHP} = \text{HHP} / 0.75 = 1552 / 0.75 = 2070$$

If an electric motor is used to drive the above pump, the actual motor horsepower required would be calculated as

Motor HP=BHP/Motor efficiency

Generally induction motors used for driving pumps have fairly high efficiencies, ranging from 95% to 98%. Using 98% for motor efficiency, we can calculate the motor HP required as follows:

$$\text{Motor HP} = 2070 / 0.98 = 2112$$

Since the closest standard-size electric motor is 2500 HP, this application will require a pump that can provide a differential pressure of 950 psi at a flow rate of 4000 bbl/hr and will be driven by a 2500 HP electric motor.

Pump companies measure pump flow rates in gal/min and pump pressures are expressed in terms of feet of liquid head. We can therefore convert the flow rate from bbl/hr to gal/min and the pump differential pressure of 950 psi can be converted to liquid head in ft.

$$4000 \text{ bbl/hr} = \frac{4000 \times 42}{60} = 2800 \text{ gal/min}$$

$$950 \text{ psi} = \frac{950 \times 2.31}{0.85} = 2582 \text{ ft}$$

The above statement for the pump requirement can then be reworded as follows: This application will use a pump that can provide a differential pressure of 2582 ft of head at 2800 gal/min and will be driven by a 2500 HP electric motor. We discuss pump performance in more detail in [Chapter 7](#).

The formula for BHP required in terms of customary pipeline units is as follows:

$$\text{BHP} = \text{QP} / (2449\text{E}) \quad (5.16)$$

where

Q=Flow rate, bbl/hr

P=Differential pressure, psi

E=Efficiency, expressed as a decimal value less than 1.0

Two additional formulas for BHP, expressed in terms of flow rate in gal/min and pressure in psi or ft of liquid, are as follows:

$$\text{BHP} = (\text{GPM})(H)(\text{Spgr}) / (3960E) \quad (5.17)$$

and

$$\text{BHP} = (\text{GPM})P / (1714E) \quad (5.18)$$

where

GPM=Flow rate, gal/min

H=Differential head, ft

P=Differential pressure, psi

E=Efficiency, expressed as a decimal value less than 1.0

Spgr=Liquid specific gravity, dimensionless

In SI units, power in kW can be calculated as follows:

$$\text{Power (kW)} = \frac{Q H \text{ Spgr}}{367.46(E)} \quad (5.19)$$

where

Q=Flow rate, m³/hr

H=Differential head, m

Spgr=Liquid specific gravity

E=Efficiency, expressed as a decimal value less than 1.0

and

$$\text{Power (kW)} = \frac{Q P}{3600(E)} \quad (5.20)$$

where

P=Pressure, kPa

Q=Flow rate, m³/hr

E=Efficiency, expressed as a decimal value less than 1.0

Example Problem 5.3

A water distribution system requires a pump that can produce 2500 ft head pressure to transport a flow rate of 5000 gal/min. Assuming a centrifugal pump driven by an electric motor, calculate the hydraulic HP, pump BHP, and motor HP required at 82% pump efficiency and 96% motor efficiency.

Solution

$$\text{Hydraulic HP} = \frac{5000 \times 62.34}{7.48} \times \frac{2500}{33,000}$$

$$\text{HHP} = 3157$$

$$\text{Pump BHP required} = \frac{3157}{0.82} = 3850 \text{ HP}$$

$$\text{Motor HP required} = \frac{3850}{0.96} = 4010 \text{ HP}$$

Example Problem 5.4

A water pipeline is used to move 320 L/s and requires a pump pressure of 750 m. Calculate the power required at 80% pump efficiency and 98% motor efficiency.

Solution

Using Equation (5.20):

$$\begin{aligned} \text{Pump power required} &= \frac{320 \times 60 \times 60}{1000} \times \frac{750 \times 1.0}{367.46 \times 0.80} \\ &= 2939 \text{ kW} \end{aligned}$$

$$\text{Motor power required} = \frac{2939}{0.98} = 3000 \text{ kW}$$

5.7 Effect of Gravity and Viscosity

It can be seen from the previous discussions that the pump BHP is directly proportional to the specific gravity of the liquid being pumped. Therefore, if the HP for pumping water is 1000, the HP required when pumping a crude oil of specific gravity 0.85 is

$$\text{Crude oil HP} = 0.85 (\text{Water HP}) = 0.85 \times 1000 = 850 \text{ HP}$$

Similarly, when pumping a liquid of specific gravity greater than 1.0, the HP required will be higher. This can be seen from examining Equation (5.17).

We can therefore conclude that, for the same pressure and flow rate, the HP required increases with the specific gravity of liquid pumped. The HP required is also affected by the viscosity of the liquid pumped. Consider water with a viscosity of 1.0 cSt. If a particular pump generates a head of 2500 ft at a flow rate of 3000 gal/min and has an efficiency of 85%, we can calculate the water HP using Equation (5.17) as follows:

$$\text{BHP} = \frac{3000 \times 2500 \times 1.0}{0.85 \times 3960} = 2228.16$$

If this pump is used with a liquid with a viscosity of 1000 SSU we must correct the pump head, flow rate, and efficiency values using the Hydraulic

Institute viscosity correction charts for centrifugal pumps. These are discussed in more detail in [Chapter 7](#). The net result is that the BHP required when pumping the high-viscosity liquid will be higher than the above calculated value. It has been found that with high-viscosity liquids the pump efficiency degrades a lot faster than flow rate or head. Also, considering pipeline hydraulics, we can say that high viscosities increase the pressure required to transport a liquid and therefore increase the HP required.

The effect of viscosity and specific gravity on pump performance will be discussed in more detail in later chapters.

5.8 System Head Curves

A system head curve, also known as a system curve, for a pipeline shows the variation of pressure required with flow rate. See Figure 5.6 for a typical system head curve. As the flow rate increases, the head required increases.

Consider a pipeline of internal diameter D and length L used to transport a liquid of specific gravity S_g and viscosity ν from a pump station at A to a delivery point at B. We can calculate the pressure required at A to transport the liquid at a particular flow rate Q . By varying flow rate Q we can determine the pressure required at A for each flow rate such that a given delivery pressure at B is maintained. For each flow rate Q , we would calculate the pressure drop due to friction for the length L of the

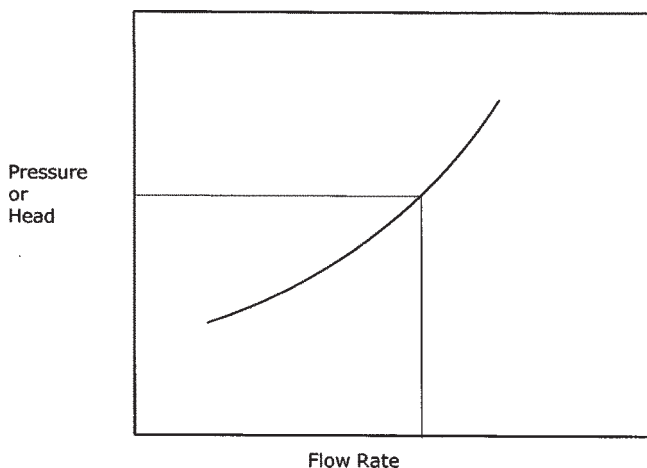


Figure 5.6 System head curve.

pipeline, add the head required to account for elevation difference between A and B, and finally add the delivery pressure required at B as follows:

$$\begin{aligned} \text{Pressure at A} &= \text{Frictional pressure drop} + \text{Elevation head} \\ &+ \text{Delivery pressure, using Equation (5.1)} \end{aligned}$$

Once the pressure at A is calculated for each flow rate we can plot a system head curve as shown in [Figure 5.6](#). The vertical axis may be in feet of liquid or psi. The horizontal axis will be in units of flow such as gal/min or bbl/hr.

In [Chapter 7](#) system head curves, along with pump head curves, will be reviewed in detail. We will see how the system head curve in conjunction with the pump head curve will determine the operating point for a particular pump-pipeline configuration.

Since the system head curve represents the pressure required to pump various flow rates through a given pipeline, we can plot a family of such curves for different liquids as shown in [Figure 5.7](#). The higher specific gravity and viscosity of diesel fuel requires greater pressures compared with gasoline. Hence the diesel system head curve is located above that of gasoline as shown in [Figure 5.7](#).

Note also that when there is no elevation difference involved the system head curve will start at the (0, 0) point. This means that at zero flow rate the pressure required is zero.

The shape of the system head curve varies depending upon the amount of friction head compared with elevation head. [Figures 5.8](#) and [5.9](#) show two system head curves that illustrate this. In [Figure 5.8](#) there is

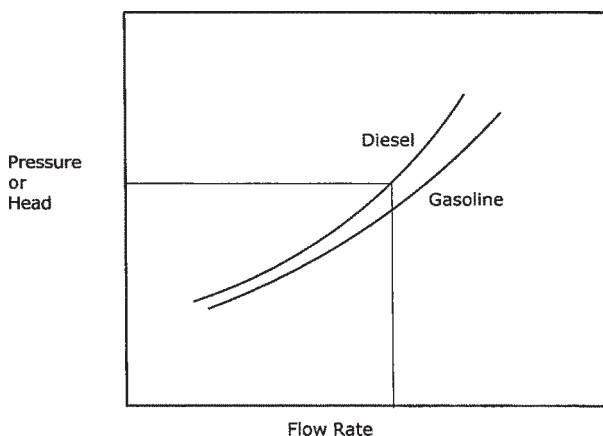


Figure 5.7 System head curve: different products.

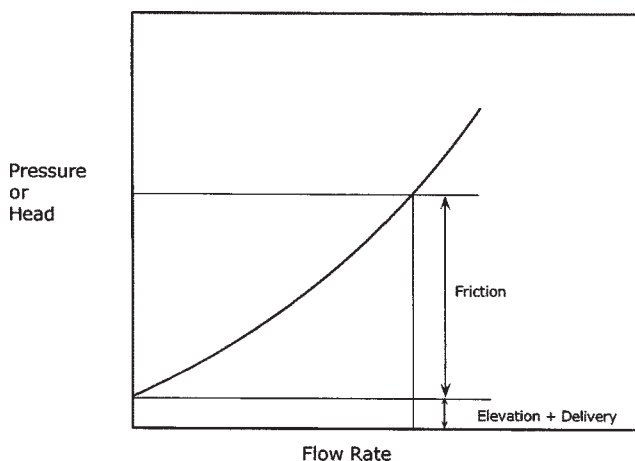


Figure 5.8 System head curve: high friction.

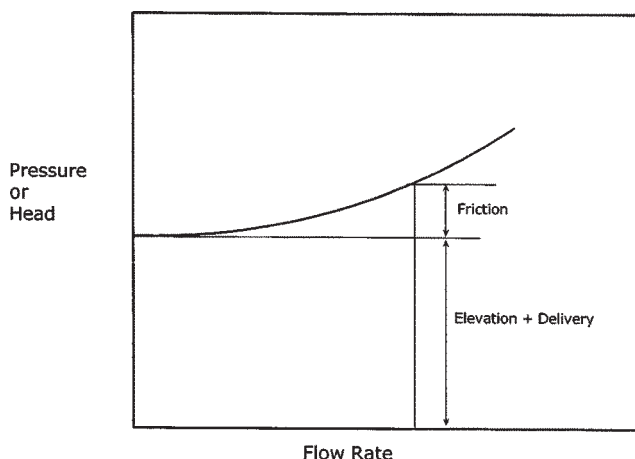


Figure 5.9 System head curve: high elevation.

comparatively less influence of the pipe elevations, most of the system head required is due to the friction in the pipe. In comparison, Figure 5.9 shows a system head curve that consists mostly of the static head due to the pipe elevations.

It can be seen from the above that the frictional head is a smaller component compared to the static elevation head.

5.9 Injections and Deliveries

In most pipelines liquid enters the pipeline at the beginning and continues to the end to be delivered at the terminus, with no deliveries or injections at any intermediate point along the pipeline. However, there are situations where liquid is delivered off the pipeline (stripping) at some intermediate location and the remainder continues to the pipeline terminus. Similarly, liquid may enter the pipeline at some intermediate location thereby adding to the existing volume in the pipeline. These are called deliveries off the pipeline and injection into the pipeline respectively. This is illustrated in Figure 5.10.

Let us analyze the pressures required in a pipeline with injection and delivery. The pipeline AB in Figure 5.10 shows 6000 bbl/hr entering the pipeline at A. At a point C, a new stream of liquid enters the pipeline at a rate of 1000 bbl/hr. Further along the pipeline, at point D, a volume of 3000 bbl/hr is stripped off the pipeline. Consequently, a resultant volume of $(6000+1000-3000)$ or 4000 bbl/hr is delivered to the pipeline terminus at B. In order to calculate the pressures required at A for such a pipeline with injection and deliveries we proceed as follows.

First, the pipe segment between A and C that has a uniform flow of 6000 bbl/hr is analyzed. The pressure drop in AC is calculated considering the 6000 bbl/hr flow rate, pipe diameter, and liquid properties. Next the pressure drop in the pipe segment CD with a flow rate of 7000 bbl/hr is calculated taking into account the blended liquid properties by combining the incoming stream at C (6000 bbl/hr) along the main line with the injection stream (1000 bbl/hr) at C. Finally, the pressure drop in the pipe segment DB is calculated considering a volume of 4000 bbl/hr and the liquid properties in that segment, which would be the same as those of pipe segment CD. The total frictional pressure drop between A and B will be the sum of the three pressure drops calculated above. After adding any elevation head between A and B and accounting for the required delivery pressure at B we can calculate the total pressure required at point A for this pipeline system. This is illustrated in the following example.

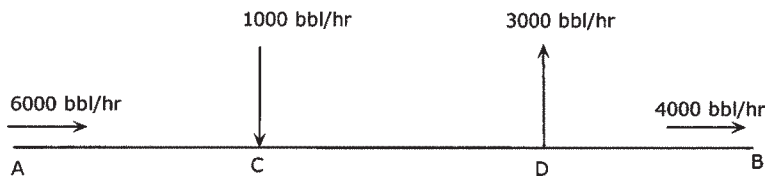


Figure 5.10 Injection and delivery.

Example Problem 5.5

In Figure 5.10 the pipeline from point A to point B is 48 miles long and is 18 in. in nominal diameter, with a 0.281 in. wall thickness. It is constructed of 5LX-65 grade steel. At A, crude oil of specific gravity 0.85 and 10 cSt viscosity enters the pipeline at a flow rate of 6000 bbl/hr. At C (milepost 22) a new stream of crude oil with a specific gravity of 0.82 and 3.5 cSt viscosity enters the pipeline at a flow rate of 1000 bbl/hr. The mixed stream then continues to point D (milepost 32) where 3000 bbl/hr is stripped off the pipeline. The remaining volume continues to the end of the pipeline at point B.

- Calculate the pressure required at A and the composition of the crude oil arriving at terminus B at a minimum delivery pressure of 50 psi. Assume elevations at A, C, D, and B to be 100, 150, 250, and 300 feet, respectively. Use the Colebrook-White equation for pressure drop calculations and assume a pipe roughness of 0.002 in.
- How much pump HP will be required to maintain this flow rate at A, assuming 50 psi pump suction pressure at A and 80% pump efficiency?
- If a positive displacement (PD) pump is used to inject the stream at C, what pressure and HP are required at C?

Solution

(a) The pressure drop due to friction for segment AC is calculated using Equation (3.27) as follows:

$$\text{Reynolds number} = 92.24 \times 6000 \times 24 / (17.438 \times 10) = 76,170$$

$$\text{Friction factor} = 0.02$$

$$\text{Pressure drop} = 13.25 \text{ psi/mile}$$

$$\text{Frictional pressure drop between A and C} = 13.25 \times 22 = 291.5 \text{ psi}$$

Next, we calculate the blended properties of the liquid stream after mixing two streams at point C, by blending 6000 bbl/hr of crude A (specific gravity of 0.85 and viscosity of 10 cSt) with 1000 bbl/hr of crude B (specific gravity of 0.82 and viscosity of 3.5 cSt) using Equations (2.4) and (2.21) as follows:

$$\text{Blended specific gravity at C} = 0.8457$$

$$\text{Blended viscosity at C} = 8.366 \text{ cSt}$$

For pipe segment CD we calculate the pressure drop by using above properties at a flow rate of 7000 bbl/hr.

$$\text{Reynolds number} = 92.24 \times 7000 \times 24 / (17.438 \times 8.366) = 106,222$$

$$\text{Friction factor} = 0.0188$$

$$\text{Pressure drop} = 16.83 \text{ psi/mile}$$

$$\text{Frictional pressure drop between C and D} = 16.83 \times 10 = 168.3 \text{ psi}$$

Finally we calculate the pressure drop for pipe segment DB by using above liquid properties at a flow rate of 4000 bbl/hr:

$$\text{Reynolds number} = 92.24 \times 4000 \times 24 / (17.438 \times 8.366) = 60,698$$

$$\text{Friction factor} = 0.021$$

$$\text{Pressure drop} = 6.13 \text{ psi/mile}$$

$$\text{Frictional pressure drop between D and B} = 6.13 \times 16 = 98.08 \text{ psi}$$

Therefore, the total frictional pressure drop between point A and point B is

$$291.5 + 168.3 + 98.08 = 557.9 \text{ psi}$$

The elevation head between A and B consists of (150-100) ft between A and C and (300-150) ft between C and B. We need to separate the total elevation head in this fashion because of differences in liquid properties in pipe segments AC and CB. Therefore, the total elevation head is

$$[(150-100) \times 0.85 / 2.31] + [(300-150) \times 0.8457 / 2.31] = 73.32 \text{ psi}$$

Adding the delivery pressure of 50 psi, the total pressure required at A is therefore

$$557.9 + 73.32 + 50 = 681.22 \text{ psi}$$

Therefore, the pressure required at A is 681.22 psi and the crude oil arriving at terminus B has a specific gravity of 0.8457 and viscosity of 8.366 cSt.

(b) The HP required at A is calculated using Equation (5.16) as follows:

$$\text{BHP} = 6000 \times (681.22 - 50) / (0.8 \times 2449) = 1933 \text{ HP}$$

(c) To calculate the injection pump requirement at point C, we must first calculate the pressure in the pipeline at point C that the PD pump has to overcome.

The pressure at C is equal to the pressure at A minus the pressure drop from A to C minus the elevation head from A to C:

$$P_C = 681.22 - 291.5 - (150 - 100) \times 0.85 / 2.31 = 371.3 \text{ psi}$$

The HP required for the PD pump at C is calculated using Equation (5.16) as follows:

$$P_D \text{ pump HP required} = (371.3 - 50) \times 1000 / (0.8 \times 2449) = 164$$

assuming 50 psi suction pressure and 80% pump efficiency.

5.10 Pipe Branches

In the previous section we discussed a pipeline with an injection point and a delivery point between the pipeline inlet and pipeline terminus. These injections and deliveries may actually consist of branch pipes bringing liquid into the main line (incoming branch) and delivering liquid out of the pipeline (outgoing branch). This is illustrated in Figure 5.11.

Sometimes we are interested in sizing the branch pipes for the required flow rates and pressures. For example, in Figure 5.11 the incoming branch BR1 needs to be sized to handle 1000 bbl/hr of liquid entering the pipeline at point C with the specified pressure P_C . We have to determine the pressure required at the beginning of branch BR1 (point E) for a specified branch pipe diameter, liquid properties, etc., so that the required pressure P_C is achieved at the end of the branch at point C.

Similarly, for an outgoing branch BR2 as shown in Figure 5.11, we are interested in determining the branch pipe size such that the liquid stream flowing through the outgoing branch arrives at its destination F with a certain specified delivery pressure. We would use the starting pressure of BR2 as P_D , which represents the main line pressure at the junction with the

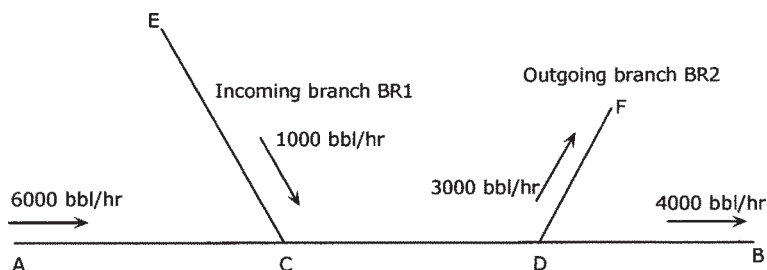


Figure 5.11 Incoming and outgoing branches.

outgoing branch. Using the value of P_D and the required delivery pressure P_F at F we can calculate the pressure drop per mile for BR2. Hence we can select a pipe size to handle the flow rate through BR2. The following example illustrates how branch pipes are sized for injection and deliveries.

Example Problem 5.6

- Using the data in Example Problem 5.5 and [Figure 5.11](#), determine the branch pipe sizes required for branch BR1 for the injection rate of 1000 bbl/hr and an MAOP of 600 psi. Assume pipe branch BR1 to be 2.5 miles long and an essentially flat elevation profile.
- Calculate the pipe size for the outgoing branch BR2 to handle the delivery of 3000 bbl/hr at D and pressure of 75 psi at F. The branch BR2 is 4 miles long.
- What HP is required at beginning of branch BR1?

Solution

(a) Assume a 6 in. pipe for branch BR1 and calculate the pressure drop in the 2.5 miles of pipe as follows: Using given liquid properties we calculate:

Pressure drop = 65.817 psi/mile for 1000 bbl/hr in a 6 in. pipe

Pressure required at E to match junction pressure of 371.3 at C is

$$P_E = 65.817 \times 2.5 + 371.3 = 535.84 \text{ psi}$$

Since this is less than the MAOP of 600 psi for BR1, a 6 in. line is adequate.

(b) With a junction pressure of 166.39 psi available at D, a 10 in. pipe is inadequate to carry 3000 bbl/hr of crude through the 4 miles of branch pipe BR2 and provide a 75 psi delivery pressure at F. Next, assuming a 12 in. pipe for BR2, 3000 bbl/hr flow rate, specific gravity of 0.8457 and viscosity of 8.366 cSt, we get

Pressure drop in BR2 = 20.07 psi/mile

Therefore, we get a delivery pressure of

$$P_F = 166.39 - 20.07 \times 4 = 86.11 \text{ psi}$$

which is higher than the minimum 50 psi delivery pressure required. Therefore, branch pipe BR2 must be at least 12 in. nominal diameter.

(c) The HP required at beginning of branch BR1 is calculated to be as follows using Equation (5.16):

$$\text{HP} = (535.84 - 550) \times 1000 / (2449 \times 0.8) = 248$$

based on 80% efficiency and 50 psi suction pressure at E.

5.11 Pipe Loops

A pipe loop is a length of parallel pipe installed between two points on a main pipeline as shown in Figure 5.12. We discussed parallel pipes and equivalent diameter earlier in this chapter. In this section we will discuss how looping an existing pipeline will reduce the pressure drop due to friction and thus require less pumping HP.

The purpose of the pipe loop is to split the flow through a parallel segment of the pipeline between the two locations, resulting in a reduced pressure drop in that segment of the pipeline.

Consider pipeline from A to B with a loop installed from C to D as shown in Figure 5.12. The flow rate between A and C is 6000 gal/min. At C where the loop is installed the flow rate of 6000 gal/min is partially diverted to the loop section with the remainder going through the mainline portion CD. If we assume that the diameter of the loop is the same as that of the main pipe, this will cause 3000 gal/min flow through the loop and an equal amount through the mainline. Assume that the section CD of the mainline prior to looping with the full flow of 6000 gal/min flowing through it had a resulting pressure drop of 25 psi/mile. With the loop installed, section CD has half the flow and therefore approximately one-fourth of the pressure drop (since the pressure drop varies as the square of the flow rate) or 6.25 psi/mile, based on Equation (3.27). If the length CD is 10 miles, the total frictional pressure drop without the loop will be 250 psi. With the pipe loop the pressure drop will be 62.5 psi, which is a significant reduction. Therefore if a pipeline section is bottlenecked due to maximum allowable operating pressure, we can reduce the overall pressure profile by installing a loop in that pipe segment.

To illustrate this concept further, consider the following example problem.

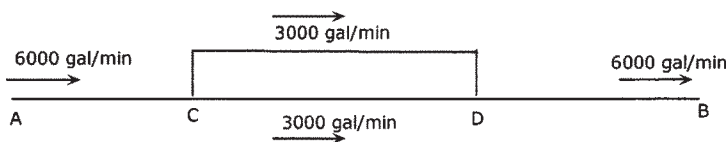


Figure 5.12 Pipe loop.

Example Problem 5.7

A 16 in. crude oil pipeline (0.250 in. wall thickness) is 30 miles long from point A to point B. The flow rate at the inlet A is 4000 bbl/hr. The crude oil properties are specific gravity of 0.85 and viscosity of 10 cSt at a flowing temperature of 70° F.

- Calculate the pressure required at A without any pipe loop. Assume 50 psi delivery pressure at the terminus B and a flat pipeline elevation profile.
- If a 10 mile portion CD, starting at milepost 10, is looped with an identical 16 in. pipeline, calculate the reduced pressure at A.
- What is the difference in pump HP required at A between cases (a) and (b) above? Assume 80% pump efficiency and 25 psi pump suction pressure.

Solution

Using the Colebrook-White equation and assuming a pipe roughness of 0.002 in., we calculate the pressure drop per unit length using Equation (3.27) as follows:

$$\text{Reynolds number} = 92.24 \times 4000 \times 24 / (15.5 \times 10) = 57,129$$

$$\text{Friction factor } f = 0.0213$$

$$\text{Pressure drop} = 11.28 \text{ psi/mile}$$

(a) Therefore the total pressure required at A = $30 \times 11.28 + 50 = 388.40$ psi without any pipe loop. With 10 miles of pipe loop the flow rate through the loop is 2000 bbl/hr and the revised pressure drop at the reduced flow is calculated as

$$\text{Pressure drop} = 3.28 \text{ psi/mile}$$

The pressure drop in the 10 mile loop = $10 \times 3.28 = 32.8$ psi

The pressure drop in the 10 mile section AC = $10 \times 11.28 = 112.8$ psi

The pressure drop in the last 10 mile section DB = $10 \times 11.28 = 112.8$ psi

Therefore

(b) Total pressure required at A = $112.8 + 32.8 + 112.8 + 50 = 308.40$ psi with the pipe loop.

Therefore, using the 10 mile loop causes a reduction of 80 psi in the pressure required at point A.

(c) HP required without the loop is

$$\text{HP} = 4000 \times (388.40 - 25) / (2449 \times 0.8) = 742$$

HP required considering the loop is

$$HP=4000 \times (308.40-25)/(2449 \times 0.8)=579$$

Thus, installing the pipe loop results in a reduction in pumping HP of
 $(742-579)/742=22\%$

What would be the impact if the looped section of pipe were smaller than the mainline? If we install a smaller pipe in parallel, the flow will still be split through the loop section but will not be equally divided. The smaller pipe will carry less flow rate than the larger mainline pipe, such that the pressure drop through the mainline from point C to point D will exactly equal the pressure drop through the pipe loop between C and D, since both the mainline pipe and the pipe loop have common pressures at the junction C and D.

To illustrate this, consider an 8 in. pipe looped with the 16 in. mainline. Assume Q_8 represents the flow rate through the 8 in. loop and Q_{16} the flow rate through the 16 in. mainline portion. We can write

$$Q_8+Q_{16}=\text{Total flow}=4000 \text{ bbl/hr} \quad (5.21)$$

From Equation (3.27), we know that the frictional pressure drop in a pipe is directly proportional to the square of the flow rate and inversely proportional to the fifth power of the diameter. Therefore, we can write

$$\text{Pressure drop in 8 in. pipe}=K(Q_8)^2/(8.625-0.5)^5$$

considering 0.250 in. wall thickness and where K is a constant,
 or

$$\Delta P_8=K(Q_8)^2/35,409 \quad (5.22)$$

Similarly, for the 16 in. pipe

$$\Delta P_{16}=K(Q_{16})^2/894,661 \quad (5.23)$$

Dividing Equation (5.22) by Equation (5.23), we get

$$\Delta P_8/\Delta P_{16}=25.27(Q_8/Q_{16})^2$$

Since ΔP_8 and ΔP_{16} are equal in a loop system, the above equation reduces to

$$Q_{16}=5.03 \times Q_8 \quad (5.24)$$

Solving Equation (5.21) and Equation (5.24) simultaneously, we get

$$Q_8=4000/6.03=664 \text{ bbl/hr} \quad \text{flow through the 8 in. loop}$$

and

$$Q_{16}=4000-664=3336 \text{ bbl/hr} \quad \begin{array}{l} \text{flow through the} \\ \text{16 in. mainline portion} \end{array}$$

We can now calculate the pressure drop and the total pressure required at A as we did in Example Problem 5.6 earlier.

Since the 10 mile looped portion of the 16 in. pipe has a flow rate of 3336 bbl/hr we calculate the pressure drop in this pipe first:

$$\text{Reynolds number}=92.24 \times 3336 \times 24 / (15.5 \times 10) = 47,646$$

$$\text{Friction factor } f = 0.0216$$

and

$$\text{Pressure drop} = 7.96 \text{ psi/mile}$$

Therefore, the pressure drop in the loop section

$$= 7.96 \times 10 = 79.6 \text{ psi}$$

The pressure drops in section AC and section DB remain the same as before. The total pressure required at A becomes

$$\text{Pressure at A} = 112.8 + 79.6 + 112.8 + 50 = 355.2 \text{ psi}$$

Compare this with 388.4 psi (without loop) and 308.4 psi (with 16 in. loop).

We could also calculate the above using the equivalent-diameter approach discussed earlier in this chapter. We will calculate the equivalent diameter of a 10 mile pipe that will replace the 8 in./16 in. loop.

Recalling our discussions of parallel pipes and equivalent diameter in Section 5.4 and using Equations (5.14) and (5.15), we calculate the equivalent diameter D_E as follows (after some simplifications):

$$D_E^{2.5} = (8.125)^{2.5} + (15.5)^{2.5}$$

$$D_E = 16.67 \text{ in.}$$

Therefore, we can say that the two parallel pipes of diameters 8 in. and 16 in. together equal a single pipe of 16.67 in. internal diameter. To show that the equivalent-diameter concept is fairly accurate, we will calculate the pressure drop in this equivalent pipe and compare it with that for the two parallel pipes.

$$\text{Reynolds number} = 92.24 \times 4000 \times 24 / (16.61 \times 10) = 53,120$$

$$\text{Friction Factor } f = 0.0211$$

and

$$\text{Pressure drop} = 7.77 \text{ psi/mile}$$

Compare this with 7.96 psi/mile we calculated earlier. It can be seen that the equivalent-diameter method is within 2% of the more exact method and is therefore accurate for most practical purposes.

A practice problem with dissimilar pipe sizes as discussed above is included at the end of this chapter as an exercise for the reader.

5.12 Summary

In this chapter we extended the pressure drop concept developed in [Chapter 3](#) to calculate the total pressure required to transport liquid through a pipeline taking into account the elevation profile of the pipeline and required delivery pressure at the terminus. We analyzed pipes in series and parallel and introduced the concept of equivalent pipe length and equivalent pipe diameter. System head curve calculations were discussed to compare pressure required at various flow rates through a pipe segment. Flow injection and delivery in pipelines were studied and the impact on the hydraulic gradient discussed. We also sized branch piping connections and discussed the advantage of looping a pipeline to reduce overall pressure drop and pumping horsepower. For a given pipeline system, the hydraulic horsepower, brake horsepower, and motor horsepower calculations were illustrated using examples. A more comprehensive analysis of pumps and HP required will be covered in [Chapter 7](#).

5.13 Problems

- 5.13.1 A pipeline 50 miles long, 16 in. outside diameter, 0.250 in. wall thickness is constructed of API 5LX-65 material. It is used to transport diesel and other refined products from the refinery at Carson to a storage tank at Compton. During phase I, a flow rate of 5000 bbl/hr of diesel fuel is to be transported with one pump station located at Carson. The required delivery pressure at Compton is 50 psi. Assume a generally rolling pipeline elevation profile without any critical peaks along the pipeline. The elevation at Carson is 100 ft, and the storage tank at Compton is located on top of a hill at an elevation of 350 ft.
- (a) Using diesel with specific gravity of 0.85 and a viscosity of 5.5 cSt at a flowing temperature of 70°F, calculate the total pressure required at Carson to transport 5000 bbl/hr of diesel on a continuous basis. Use the Hazen-Williams equation with a C-factor of 125. The Carson pump

suction pressure is 30 psi and the pump is driven by a constant-speed electric motor.

- (b) Determine the BHP and motor HP required at Carson assuming 82% pump efficiency and 96% motor efficiency.
 - (c) What size electric motor would be required at Carson?
 - (d) Assuming a maximum allowable operating pressure of 1400 psi for the pipeline, how much additional throughput can be achieved in phase II if the pumps are modified at Carson?
- 5.13.2 A water pipeline is being built to transport 2300 m³/hr from a storage tank at Lyon (elevation 500m) to a distribution facility 50 km away at Fenner (elevation 850 m). What size pipe will be required to limit velocities to 3 m/s and allowable pipeline pressure of 5.5MPa? No intermediate pump stations are to be used. Delivery pressure at Fenner is 0.3 MPa. Use the Hazen-Williams equation with a C-factor of 110.
- 5.13.3 In Problem 5.13.1, what throughputs can be achieved when pumping gasoline alone? Use a specific gravity of 0.74 and viscosity of 0.65 for gasoline at the flowing temperature. Compare the pump head requirements when pumping diesel versus gasoline. Use the Hazen-Williams equation with a C-factor of 145.
- 5.13.4 Calculate the BHP requirements for the above problem for both diesel and gasoline movements.
- 5.13.5 Consider a loop pipeline similar to that in Example Problem 5.7. Instead of the loop being 16 in. consider a smaller 10 in. pipe installed in parallel for the middle 10 mile section. How does the pressure at A change compared with the no-loop case?